

# STRESSPREDICT: A Questionnaire-Based Intelligent System for Stress Level Prediction Using Machine Learning Inspired Techniques

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## Abstract

Stress has become one of the most critical challenges in modern society due to increasing academic pressure, professional workload, and lifestyle imbalance. Early detection of stress is essential to prevent long-term mental and physical health issues. Traditional stress detection techniques rely on physiological signals such as heart rate variability, EEG, and wearable sensors, which are expensive and not accessible to everyone.

This paper presents **STRESSPREDICT**, a questionnaire-based stress prediction system inspired by machine learning concepts. The system evaluates user responses related to psychological and behavioural factors such as sleep quality, workload, anxiety, emotional state, and concentration. A weighted scoring mechanism is applied to compute a stress score and classify it into Low, Moderate, or High stress levels. The system focuses on **simplicity, explainability, and real-time usability**, making it practical for everyday users without requiring datasets or hardware. Additionally, graphical visualization is integrated to enhance user understanding.

## 1. Introduction

Stress is a natural psychological response to demanding situations, but excessive stress can negatively affect mental health, productivity, and overall well-being. In today's fast-paced environment, students and professionals frequently experience stress due to deadlines, academic pressure, multitasking, and lifestyle imbalance.

Traditional stress detection methods involve physiological monitoring such as heart rate variability, electroencephalography (EEG), and skin conductance. While these approaches provide accurate results, they require specialized equipment and controlled environments, making them impractical for general use.

Machine learning-based approaches have improved stress prediction accuracy by analysing large datasets. However, these methods require:

- Large labelled datasets
- Training and testing phases
- High computational resources

Therefore, there is a need for a **simple, accessible, and explainable system** that can predict stress without requiring complex infrastructure. The proposed system addresses this gap using a questionnaire-based approach combined with machine learning-inspired logic.

## 2. Literature Review

Several approaches have been proposed for stress detection:

### Physiological-Based Methods

These methods use biological signals such as heart rate and EEG. Although accurate, they require expensive hardware and are not suitable for everyday users.

### Machine Learning Models

Algorithms such as Decision Trees, Random Forest, and Support Vector Machines are widely used for stress classification. These models improve prediction accuracy but depend on datasets and training.

### Wearable Device-Based Systems

Smart devices collect real-time physiological data, but their effectiveness depends on device availability and battery life.

### Questionnaire-Based Systems

Psychological studies commonly use questionnaires for stress assessment. These systems are:

- Cost-effective
- Easy to implement
- User-friendly

Recent research such as **"Explainable Machine Learning for Multiclass Stress Prediction Using Questionnaire Data (IEEE Access, 2026)"**

focuses on interpretability and transparency.

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### Research Gap

Most existing systems either:

- Require hardware
- Are complex and non-explainable
- Are not suitable for real-time use

This project aims to bridge these gaps.

## 3. Proposed System

The proposed system is a modular stress prediction model consisting of:

1. **Input Module** – Collects user responses
2. **Feature Encoding** – Converts responses into numerical values
3. **Scoring Engine** – Calculates stress score

4. **Classification Module** – Determines stress level
5. **Explanation Module** – Identifies key stress factors
6. **Suggestion Module** – Provides recommendations
7. **Visualization Module** – Displays graphical output

#### 4. Methodology

##### 4.1 Data Representation

User responses are encoded as:

- No → 0
- Sometimes → 1
- Often → 2

These values act as input features.

##### 4.2 Weighted Scoring Model

The stress score is calculated using:

$$S = \sum_{i=1}^n w_i \cdot x_i$$

Where:

- $x_i$  = user response
- $w_i$  = weight of factor
- $S$  = stress score

Weights are assigned based on importance:

- Sleep → high weight
- Workload → high weight
- Anxiety → moderate weight

##### 4.3 Classification

$$\text{Stress Level} = \begin{cases} \text{Low} & S < 10 \\ \text{Moderate} & 10 \leq S < 20 \\ \text{High} & S \geq 20 \end{cases}$$

##### 4.4 Explainability

The system identifies major contributing factors such as:

- Poor sleep
- Lack of concentration
- High workload

This makes the system **transparent and interpretable**.



#### 5. Implementation

The system is implemented using **Python**.

##### Libraries Used

- matplotlib (for visualization)

##### Workflow

1. User inputs answers
2. Responses are processed
3. Score is calculated
4. Stress level is classified
5. Reasons are displayed
6. Suggestions are generated
7. Graph is plotted

#### 6. Results and Analysis

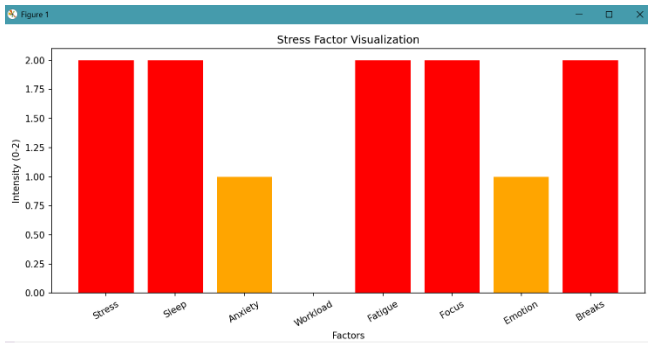
The system successfully:

- Calculates stress score
- Classifies stress level
- Identifies key contributing factors
- Provides graphical representation

Example:

- Score: 24
- Level: High

Graph visualization helps users understand which factors contribute most to stress.



Hello! Let's understand how you're feeling today.  
Please answer honestly (0 = No, 1 = Sometimes, 2 = Often)

1. Do you feel mentally overwhelmed? (0-2): 2
2. How would you rate your sleep quality? (0-2): 2
3. Do you feel anxious or restless? (0-2): 2
4. Are you dealing with heavy workload or pressure? (0-2): 2
5. Do you feel physically or mentally tired? (0-2): 2
6. Do you find it hard to concentrate? (0-2): 2
7. Do you feel emotionally drained? (0-2): 2
8. Do you take enough breaks or relaxation time? (0-2): 2

Analysis Result:

It seems like you're experiencing high stress.

Key Factors:

- high workload pressure
- increased anxiety levels
- lack of concentration
- emotional exhaustion

Suggestions:

- Take proper rest
- Try meditation or breathing exercises
- Reduce workload
- Talk to someone you trust

⚠️ This is not a medical diagnosis.

Stress Score: 38  
Stress Level: High

- Integration with machine learning models
- Mobile or web application
- AI chatbot integration
- Dataset-based improvement
- Real-time monitoring

## 11. Conclusion

The proposed system demonstrates that effective stress prediction can be achieved using simple and explainable methods. By combining questionnaire-based input with machine learning-inspired logic, the system provides a practical solution for everyday stress monitoring.

## References

1. M. Sharisha Shanbhog, Jeevan Medikonda, Shweta Rai, and Rayan Charls Mathias, *Explainable Machine Learning for Multiclass Stress Prediction Using Questionnaire Data*, IEEE Access, 2026.

## 7. Novelty of the Project

The key innovations of this system include:

- **Hybrid Approach:** Combines questionnaire with ML concepts
- **Explainable Output:** Provides reasons behind results
- **Weighted Scoring:** More accurate than simple counting
- **Real-Time Analysis:** No training required
- **Graph Visualization:** Enhances understanding
- **No Dataset Required:** Works independently

## 8. Advantages

- Easy to use
- Cost-effective
- No hardware required
- Explainable results
- Fast execution

## 9. Limitations

- Depends on user honesty
- Not clinically validated
- Limited ML complexity

## 10. Future Scope